

SHWETAV DE

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PROFESSIONAL SUMMARY

Final-year B.Tech Computer Science & Engineering student with hands-on experience in building AI-powered applications, including a conversational RAG pipeline for YouTube video Q&A, a voice-activated virtual assistant using OpenAI APIs, and a real-time weather web application. Proficient in Python, JavaScript, SQL, and Machine Learning, with exposure to LangChain, HuggingFace, FAISS, and cloud platforms (AWS, IBM Cloud). Actively seeking a Software Engineer or AI/ML role to contribute effectively within a dynamic team.

TECHNICAL SKILLS

Languages: Python, C, C++, JavaScript, HTML, CSS, PHP

Databases: MySQL

AI / ML & NLP: LangChain, HuggingFace Transformers, FAISS, RAG, OpenAI API (GPT-3.5-turbo, Arcee-Trinity), Generative AI, Machine Learning, Data Preprocessing, Clustering (K-Means)

PROJECTS

YouTube Video Q&A — Conversational RAG System | *Python · LangChain · HuggingFace · FAISS*

- Built an end-to-end conversational RAG pipeline that ingests YouTube video transcripts and answers user questions strictly grounded in video content, preventing hallucination.
- Implemented a two-chain architecture: a condense chain to rephrase follow-up questions into standalone queries, and a RAG chain to retrieve relevant transcript chunks and generate accurate answers.
- Used HuggingFaceEmbeddings and FAISS vectorstore for semantic chunk retrieval, and LangChain's RecursiveCharacterTextSplitter for optimal document chunking.
- Maintained full multi-turn conversational context using LangChain ChatPromptTemplate, MessagesPlaceholder, and HumanMessage/AIMessage history tracking.

Jarvis — AI Voice-Activated Virtual Assistant | *Python · OpenAI API*

- Engineered a fully functional voice assistant capable of web browsing, music playback, real-time news fetching, and natural language Q&A using OpenAI's Arcee-Trinity LLM.
- Implemented speech recognition and text-to-speech pipelines, enabling end-to-end voice-driven interaction with fully automated processing.
- Demonstrated applied knowledge of Generative AI integration, API consumption, and event-driven programming in Python.

ReplyGenie — AI-Powered Auto-Reply Chatbot | *Python · GPT-3.5-turbo*

- Built an intelligent chatbot that analyses conversation history and auto-generates contextual, persona-driven responses using OpenAI's GPT-3.5-turbo API.
- Designed a custom AI persona with consistent tone and style, showcasing prompt engineering and LLM customisation skills.
- Automated a full chat-interaction pipeline — from history parsing to response generation — reducing manual effort to zero.

Weather Application | *HTML · CSS · JavaScript · REST API*

- Developed a responsive, mobile-friendly web application that fetches and displays real-time weather data (temperature, humidity, forecast) via a public weather REST API.
- Implemented dynamic UI updates and error-handling logic for invalid locations, ensuring a smooth user experience across devices.

Customer Segmentation — Machine Learning | *Python · Scikit-learn*

- Applied K-Means clustering and data preprocessing techniques to segment customers based on purchasing behaviour patterns.
- Cleaned and transformed raw datasets, performed EDA, and visualised cluster distributions to derive actionable business insights.

EDUCATION

B.Tech — Computer Science & Business Systems

Sept 2022 – Jun 2026 (Expected)

Techno Main Saltlake, Kolkata | CGPA: 7.0 / 10

- Relevant coursework: Data Structures & Algorithms, Database Management Systems, Machine Learning, Operating Systems, Computer Networks.

Senior Secondary (Class XII): 90% | Secondary (Class X): 86.28%

CERTIFICATIONS & COURSES

- Introduction to Machine Learning — NPTEL, IIT Madras (Jul – Oct 2025)